Ablation studies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Ablation studies before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Multiple comparisons

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Effect size and practical importance

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Confounders and controls

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

Complete the supported Ablation studies trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Ablation** studies. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Ablation studies explanation against the supplied facts before discussing it.

Use the worked case for **Ablation** studies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Full12/15, without A9/15, without B11/15. Compute sample drops and state **interaction** limit.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Ablation studies facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Ablation studies facts and keep your reasoning.

Full system8/10, without retrieval5/10, without validation7/10; the last permits an unauthorised action. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Ablation studies facts and keep your reasoning.

A report omits this qualification: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Ablation studies facts and keep your reasoning.

Full12/15, without A9/15, without B11/15. Compute sample drops and state **interaction** limit.

> **My reasoning and evidence**

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Ablation studies facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Ablation studies to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled **ablation** studies **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **ablation** studies.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Ablation studies

> Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter’s contribution. Explain the labelled step before checking the changed condition.

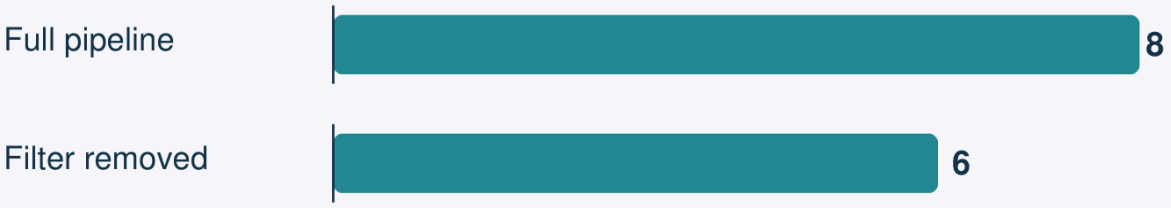
Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter’s contribution.

Full pipeline: 8/10 correct. Filter removed: 6/10 on the same cases, with all other settings fixed.

AI ACADEMY / AI-7/16

Relevance-filter ablation

correct answers out of 10 shared cases



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/16

Ablation studies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the component

The relevance filter is the single declared change.

↓



2. Run the ablation

Remove only the filter and preserve cases and scoring.

↓



3. Compare outcomes

Correctness falls from 80% to 60%, two cases in this set.

CHANGE ONE THING

Remove the filter and also change the model version.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate the filter **ablation**'s two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Correctness falls from **80%** to **60%**, two cases in this set.

1. Name the component

2. Run the **ablation**

3. Compare outcomes

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare outcomes Correctness falls from **80%** to **60%**, two cases in this set. An **ablation** probes a component by removing or changing it while holding the rest fixed.

It supports a scoped comparison, not a claim that the component always helps.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove the filter and also change the model version. Which conclusion is supported by the supplied facts?

1. The comparison no longer isolates the filter.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The comparison no longer

isolates the filter. Two simultaneous changes provide alternative explanations.

Repair a missing building block

Review Multiple comparisons (ai-7/15). Goal: Explain the **mechanism** and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Review Effect size and practical importance (ai-7/13). Goal: Explain the **mechanism** and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000.

Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Ablation studies picture from its labels, then record a relevant limit.

01. Ablation studies: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

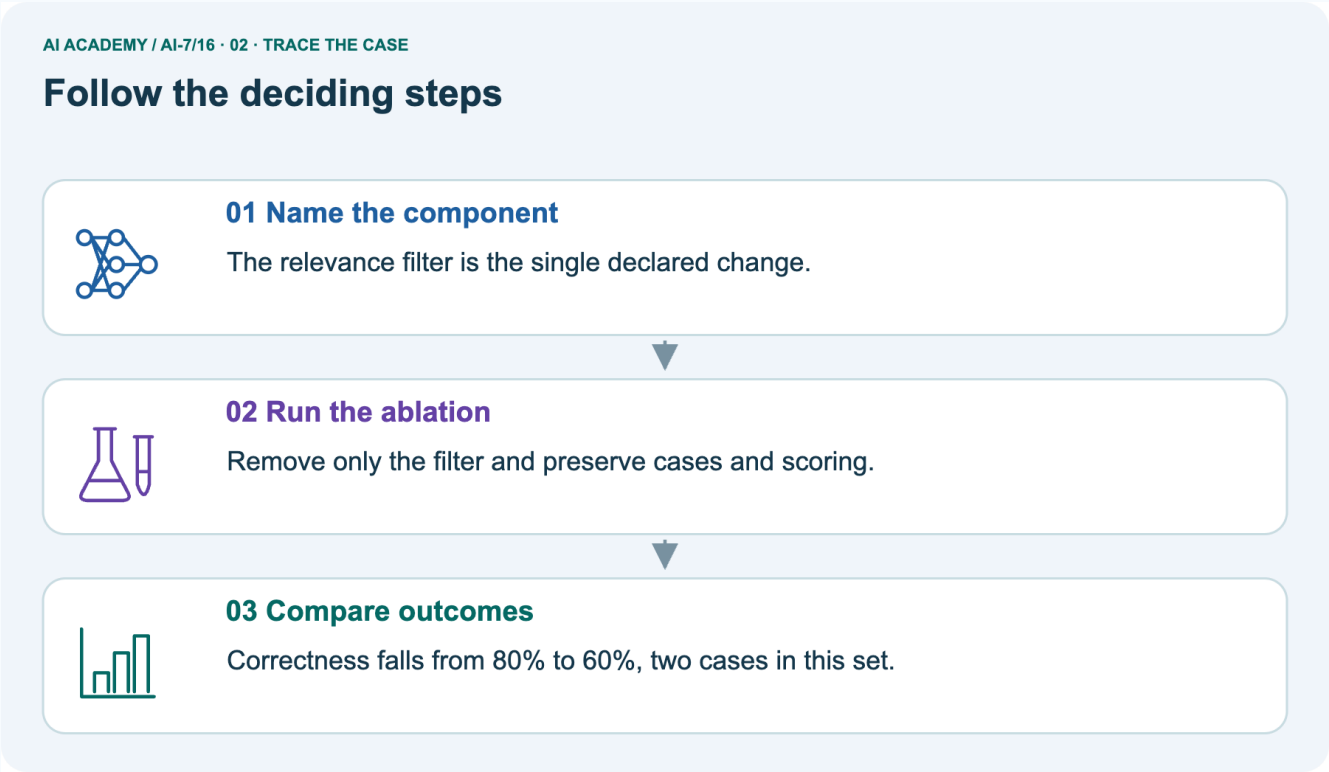
Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution. Supplied facts: Full pipeline: 8/10 correct.

Filter removed: 6/10 on the same cases, with all other settings fixed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



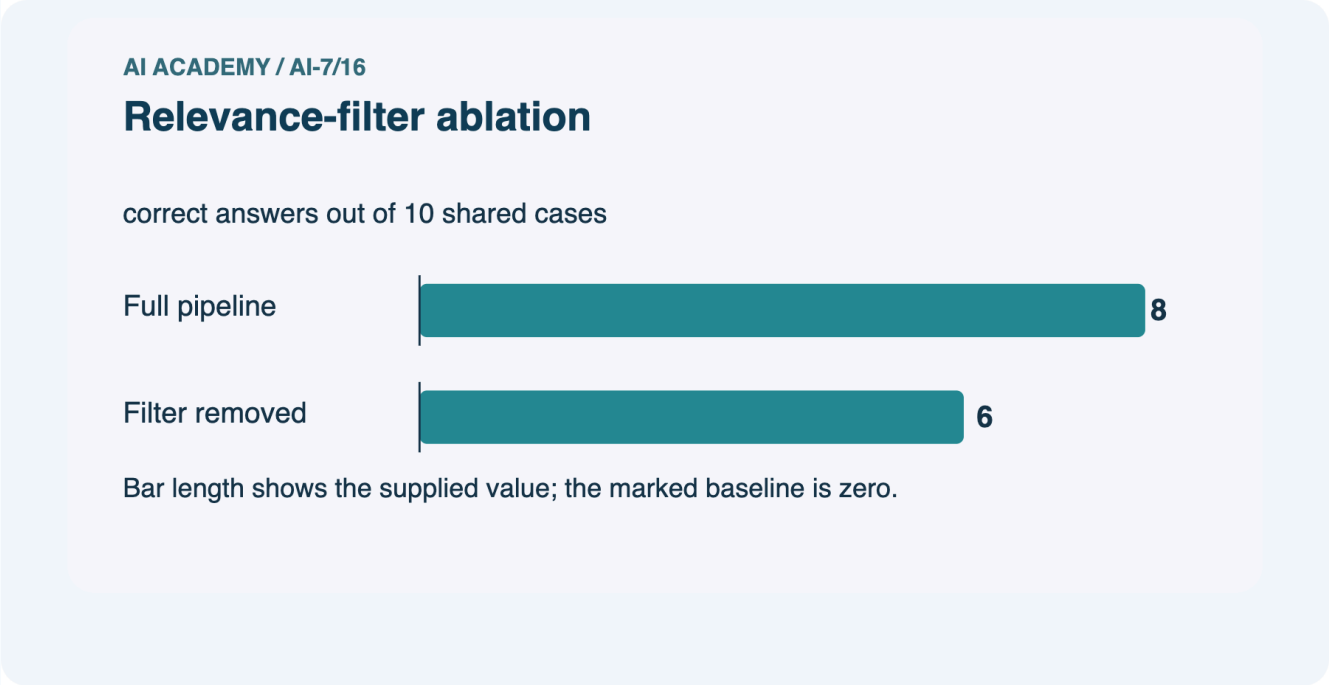
Follow the labelled picture with the matching explanation. Open the vector diagram

Name the component: The relevance filter is the single declared change. → Run the **ablation**: Remove only the filter and preserve cases and scoring. → Compare outcomes: Correctness falls from **80%** to **60%**, two cases in this set.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Relevance-filter ablation



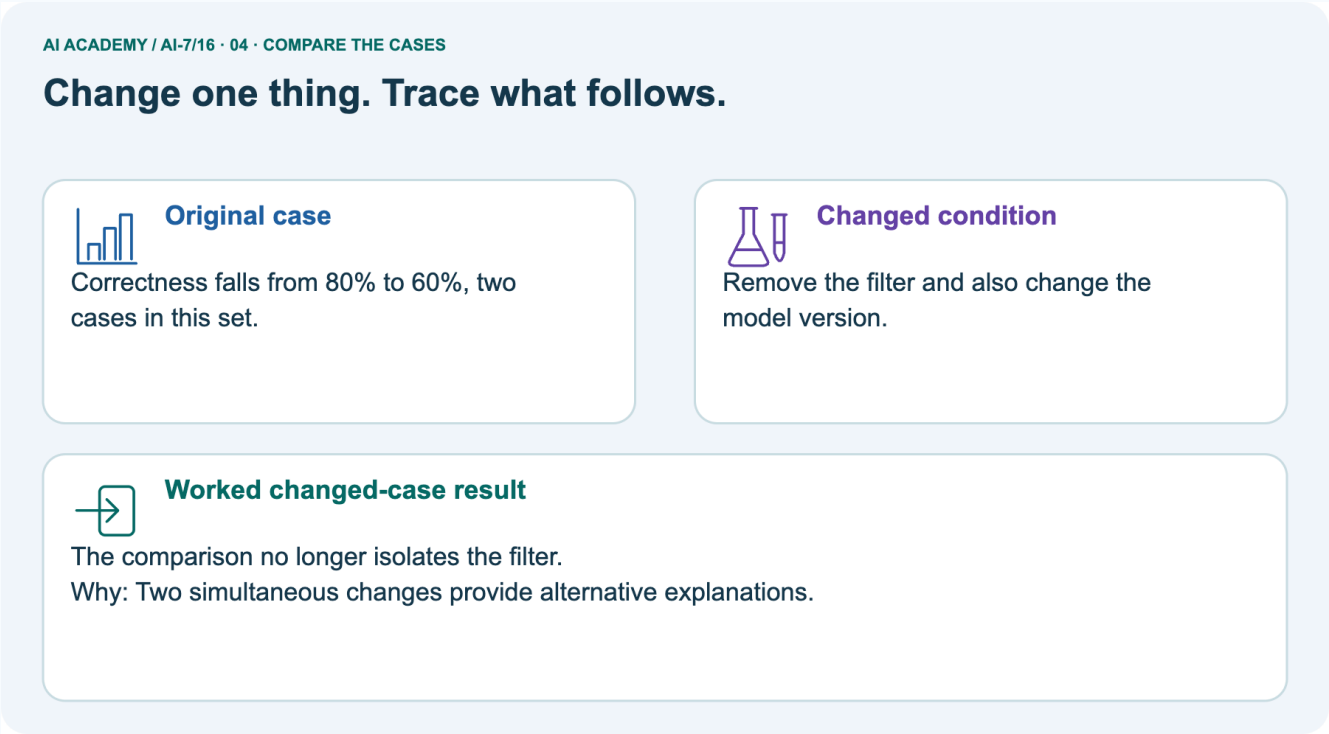
Follow the labelled picture with the matching explanation. Open the vector diagram

Relevance-filter **ablation**. Units: correct answers out of 10 shared cases. Full pipeline: 8. Filter removed: 6.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

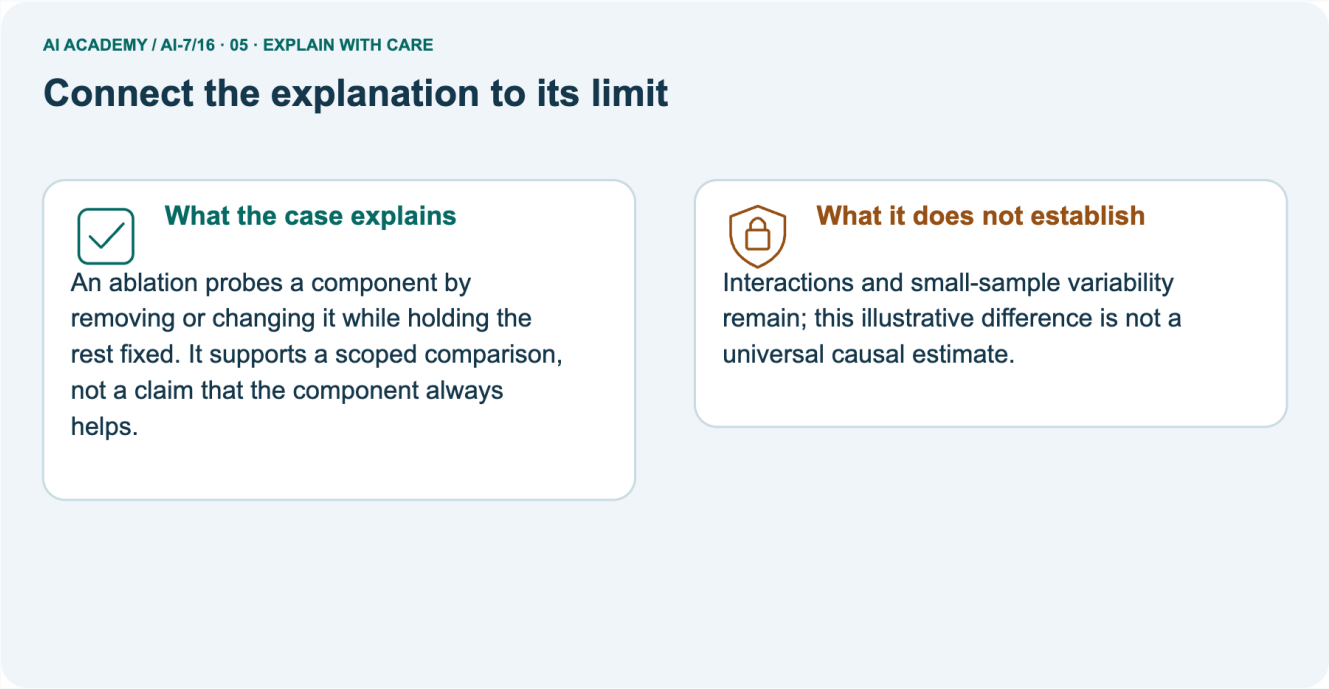
Original case: Correctness falls from **80%** to **60%**, two cases in this set. Changed condition: Remove the filter and also change the model version. Expected result: The comparison no longer isolates the filter.

Reason: Two simultaneous changes provide alternative explanations.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

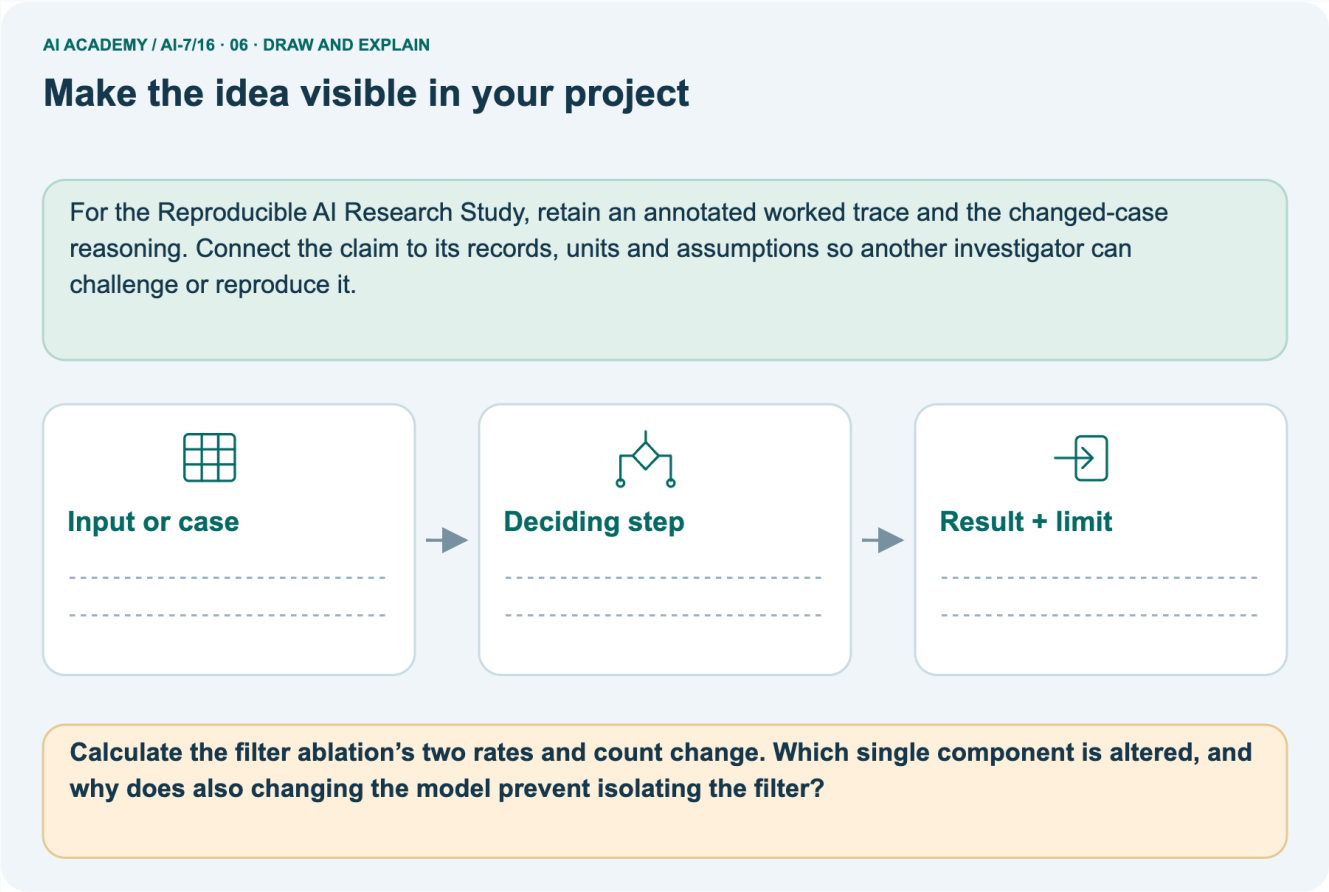
An **ablation** probes a component by removing or changing it while holding the rest fixed. It supports a scoped comparison, not a claim that the component always helps.

Limit: **Interaction**s and small-sample variability remain; this illustrative difference is not a universal causal estimate.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the filter **ablation**’s two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.